ECM-402 Lab 7 Report

Unsupervised Learning: Independent Component Analysis, PCA and Clustering

Blind Source Separation, Dimensionality Reduction, and Cluster Validity

Abstract

This report presents an exhaustive theoretical, mathematical, and empirical investigation of unsupervised learning applied to signal processing data. Three techniques are studied, each answering a different question about unlabelled measurements: Independent Component Analysis (ICA), which asks which independent sources were mixed to produce the observed channels; Principal Component Analysis (PCA), which asks which few directions carry most of the variance; and k-means together with its alternatives, which ask which measurements group together. The through-line of the study is a distinction that is made quantitative throughout: PCA removes *correlation*, a second-order property, whereas ICA removes statistical *dependence*, a property of all orders.

All data are synthetic and generated by code, so the ground truth is known. It is used strictly for evaluation (adjusted Rand index, normalised mutual information, contingency tables, matched correlation) or for colouring plots, and never for fitting a model or choosing a hyperparameter. First, a three-channel instantaneous mixture of a sinusoid, a square wave and a sawtooth wave (condition number $\mathrm{cond}(A) = 9.85$) is centred, whitened to $\mathrm{cov}(X_w) = I_3$ within 1.3×10^{-14}, and unmixed by FastICA with a mean matched absolute correlation of 0.99968, compared with only 0.702 for PCA. A squared-component covariance statistic D_{ij} confirms that PCA outputs are uncorrelated yet dependent (mean off-diagonal $|D_{ij}| = 0.450$) while ICA outputs are near the level of the true sources (0.0124 against 0.0065). Replacing the sources with Gaussian signals removes identifiability: FastICA scores 0.807, inside a random-rotation baseline of 0.827 (5th–95th percentile 0.694–0.955).

Second, PCA is applied to twelve higher-order-cumulant and amplitude features of a synthetic four-class modulation dataset (BPSK, QPSK, 8PSK, 16QAM at 20 dB). Three components retain 90% of the variance and five retain 95%; the reconstruction error equals the discarded eigenvalue mass to 10^{-15}, verifying least-squares optimality. Third, k-means at $k = 4$ recovers the classes with ARI $= 0.9973$, yet the silhouette score prefers $k = 3$ because it merges the closely spaced QPSK and 8PSK classes, a concrete failure of an internal index. The dependence of clustering quality on SNR is measured over ten independent datasets. Fourth, GMM and Ward clustering are compared with k-means, and a two-moons experiment shows k-means reaching ARI $= 0.257$ where spectral clustering reaches 1.0, delineating exactly when centroid-based clustering is the wrong tool.

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Part I

Blind Source Separation with Independent Component Analysis

1 Experimental Protocol and Reproducibility

1.1 Software and Execution Environment

Unsupervised learning has a trap that supervised learning does not: it is tempting to peek at the labels while tuning. Every experiment in this report is therefore executed under a fixed protocol. All computations use `numpy`, `scipy`, `scikit-learn` and `matplotlib` on a CPU, with no GPU and no deep-learning framework. The full set of figures and tables is regenerated by a single entry script, `python src/main.py`, which calls separate modules for blind source separation (Part I), dataset and feature generation and PCA (Part II), and clustering (Parts III and IV).

1.2 Random Seeds

Unsupervised results can vary noticeably between runs, and k-means in particular depends on initialisation. Every source of randomness is therefore fixed and reported (Table 1). All sample covariance matrices use the $N-1$ denominator throughout, so that eigenvalue-based identities such as the reconstruction-error formula of Section 9 hold without a stray factor.

Component	**Seed / setting**
Source waveforms and noise	`np.random.seed(0)`
Gaussian sources	`default_rng(1)`
Random-rotation baseline	`default_rng(123)`, 200 rotations
FastICA	`random_state=0`, `whiten="unit-variance"`, `max_iter=1000`
Modulation dataset	`seed=0`, 250 signals per class, 256 symbols
KMeans	`random_state=0`, `n_init=10`
GaussianMixture, SpectralClustering	`random_state=0`
`make_moons`	`n_samples=500`, `noise=0.05`, `random_state=0`

Table 1: Random seeds and solver settings used for every experiment.

1.3 Label Policy

The ground-truth labels are used under three explicit rules:

- **Fit on features alone.** No label is ever passed to a `fit()` call of an unsupervised model.
- **Select with internal criteria only.** The number of clusters k and the number of principal components are chosen with internal quantities (inertia, silhouette, explained variance), never with a label-based score.
- **Evaluate and colour with labels.** Labels appear only in external metrics (ARI, NMI, contingency table, matched correlation) or to colour plots. The single exception is the supervised stand-in classifier of Section 14, which is supervised by definition.

In a real deployment these labels would not exist; here they are a teaching aid that allows us to check whether each method worked.

2 Linear Mixing Model and Source Construction

2.1 The Instantaneous Mixing Model

Suppose three microphones record three simultaneous sound sources. In the simplest instantaneous model, each microphone records a different linear combination of the sources:

$$\mathbf{x}(t) = A\,\mathbf{s}(t), \qquad \mathbf{x}(t), \mathbf{s}(t) \in \mathbb{R}^3, \quad A \in \mathbb{R}^{3\times 3}, \tag{1}$$

where $\mathbf{s}(t)$ holds the unknown sources and A is the unknown mixing matrix. The *blind source separation* (BSS) problem is to recover the sources from the mixtures alone. ICA solves it by estimating an unmixing matrix W such that

$$\hat{\mathbf{s}}(t) = W\,\mathbf{x}(t) \tag{2}$$

has components that are as statistically independent as possible. The standard ICA model assumes that the sources are mutually independent and that at most one of them is Gaussian. Because neither the sources nor A are known, three ambiguities are unavoidable: the *order* of the components, their *scale*, and their *sign* cannot be determined, so a recovered source may appear permuted, rescaled or inverted relative to the true one.

2.2 Source Signals and Mixtures

Three illustrative sources are generated on $t \in [0, 8]$ with $N = 2000$ samples: a sinusoid of frequency 1.0 (source s_1), a square wave of frequency 0.7 (s_2) and a sawtooth of frequency 1.3 (s_3). Small Gaussian noise of standard deviation 0.05 is added and each column is scaled to unit standard deviation. The mixing matrix is

$$A = \begin{pmatrix} 1.0 & 1.0 & 0.6 \\ 0.6 & 1.8 & 1.0 \\ 1.4 & 0.7 & 1.3 \end{pmatrix}, \qquad X = SA^{\top}. \tag{3}$$

These waveforms are illustrative: periodic signals with different frequencies are only approximately independent, so the experiment should not be read as a rigorous construction of independent sources.

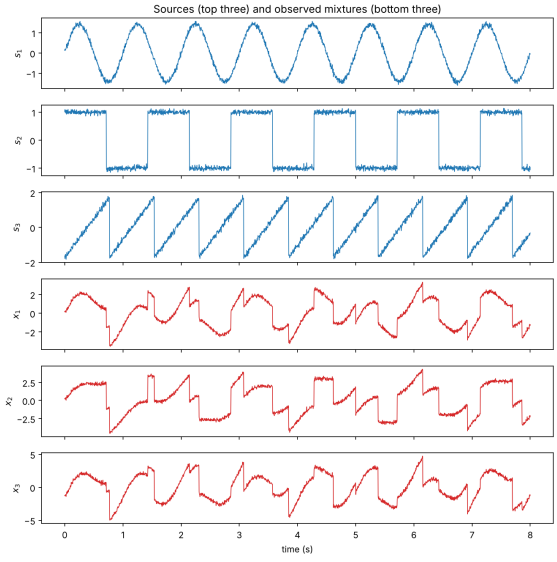


Figure 1: Three source signals (top) and the three observed mixtures (bottom) on a shared time axis.

Figure 1 shows that every mixture is a weighted sum of all three waveforms, so each x_i shows the sine, square and sawtooth superimposed. The mixing weights differ by channel and are unknown, so no mixture resembles any single source, and the sources cannot be read off by visual inspection.

2.3 Conditioning of the Mixing Matrix

The mixing matrix has singular values 3.167, 0.982 and 0.321 and determinant $\det A = 1.0$, giving

$$\mathrm{cond}(A) = \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{3.167}{0.321} = 9.85. \tag{4}$$

Since A has full rank it is invertible, so exact recovery is possible in principle. The condition number controls how fragile that recovery is. If the observations are perturbed by an error $\mathbf{e}$, the unmixing error obeys

$$\frac{\|A^{-1}\mathbf{e}\|}{\|A^{-1}\mathbf{x}\|} \;\le\; \mathrm{cond}(A)\,\frac{\|\mathbf{e}\|}{\|\mathbf{x}\|}, \tag{5}$$

so noise or estimation error along the weakest singular direction is amplified by up to $\mathrm{cond}(A)$. A very large condition number means that the columns of A are nearly linearly dependent: two sources then produce almost the same pattern across the channels, and separating them requires dividing by a tiny singular value. In the limit $\mathrm{cond}(A) \to \infty$ the matrix becomes singular, some linear combination of the sources is invisible to the sensors, and the corresponding sources are not recoverable at all. Here $\mathrm{cond}(A) = 9.85$ is mild, so the problem is well posed.

3 Centering and Whitening

3.1 Whitening Transform

ICA algorithms commonly centre and whiten the observations first. Let X_c be the centred data matrix and let the sample covariance have the eigendecomposition

$$\Sigma = E\Lambda E^{\top}. \tag{6}$$

The whitened data are

$$X_w = X_c\, E\, \Lambda^{-1/2}. \tag{7}$$

The eigenvalues of $\mathrm{cov}(X)$ are 0.107, 0.915 and 10.242, so the strongest direction carries almost all of the signal energy. After whitening, $\mathrm{cov}(X_w) = I_3$ to within 1.3×10^{-14}: the off-diagonal entries are below 10^{-15} and the diagonal entries equal 1.0000000000000133, 1.0000000000000004 and 1.0000000000000004. The deviation is pure floating-point round-off.

3.2 Why Whitening Simplifies the Problem

Whitening removes all second-order structure, namely the scale and the correlation of the channels. To see what remains, let $V = \Lambda^{-1/2}E^{\top}$ be the whitening matrix and let the sources have identity covariance. Then

$$\mathrm{cov}(V\mathbf{x}) = VAA^{\top}V^{\top} = I \implies (VA)(VA)^{\top} = I, \tag{8}$$

so VA is an orthogonal matrix. In the ideal square, noiseless, full-rank model, ICA only has to search for a rotation. An orthogonal $n \times n$ matrix has $n(n-1)/2$ independent degrees of freedom, whereas a general matrix has n^2 (Table 2). For $n = 3$ the search space shrinks from 9 to 3 parameters; the other 6 are already fixed by the whitening step. This lets FastICA use fast orthogonal updates and avoids the ill-conditioning of an unconstrained search.

Components n	General matrix n^2	Orthogonal $n(n-1)/2$	Fixed by whitening
2	4	1	3
3	9	3	6
5	25	10	15
10	100	45	55

Table 2: Degrees of freedom of a general unmixing matrix versus an orthogonal one. Whitening removes $n(n+1)/2$ parameters.

4 FastICA Separation and Evaluation

4.1 The FastICA Principle

FastICA estimates each unmixing direction $\mathbf{w}$ by maximising the non-Gaussianity of $\mathbf{w}^\top \mathbf{z}$ for whitened data $\mathbf{z}$. Non-Gaussianity is measured by an approximation of negentropy,

$$J(y) \;\propto\; \big[\mathbb{E}\{G(y)\} - \mathbb{E}\{G(\nu)\}\big]^2, \qquad \nu \sim \mathcal{N}(0,1), \tag{9}$$

with the default contrast $G(u) = \log\cosh u$ of `scikit-learn`. Writing $g = G'$, the fixed-point iteration is

$$\mathbf{w} \leftarrow \mathbb{E}\{\mathbf{z}\, g(\mathbf{w}^\top \mathbf{z})\} - \mathbb{E}\{g'(\mathbf{w}^\top \mathbf{z})\}\, \mathbf{w}, \qquad \mathbf{w} \leftarrow \frac{\mathbf{w}}{\|\mathbf{w}\|}, \tag{10}$$

and in the parallel variant used here all directions are updated together and decorrelated by $W \leftarrow (WW^\top)^{-1/2} W$. The algorithm was run on the observed mixtures only: neither S nor A was supplied. It converged in 7 iterations with no warnings.

4.2 Recovered Components and Matching

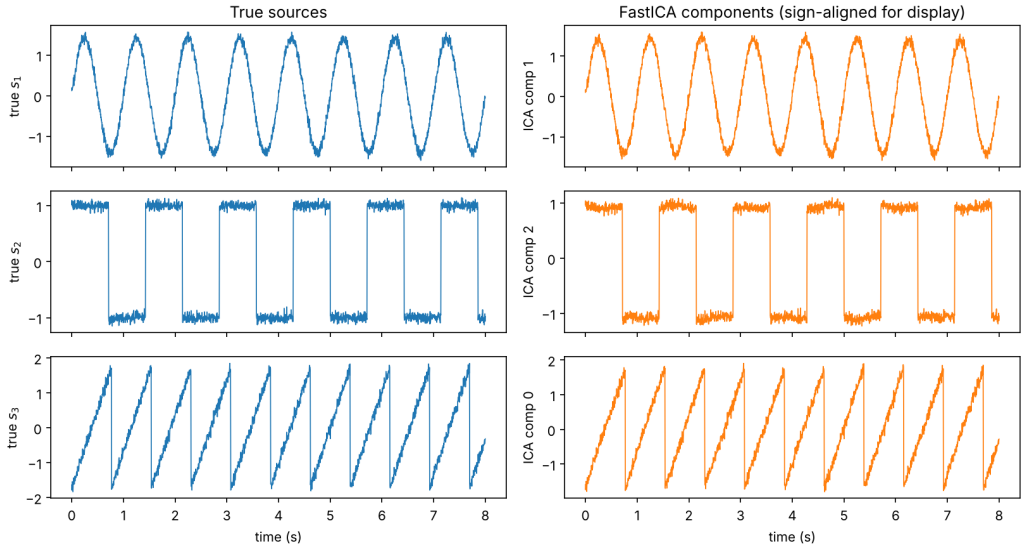


Figure 2: True sources (left) and FastICA components (right). Signs are aligned for display only.

Figure 2 compares the true sources with the recovered components. Because of the permutation and sign ambiguity, components cannot be compared index by index. Instead, the 3×3 matrix of absolute Pearson correlations is computed (rows s_1, s_2, s_3, columns IC0, IC1, IC2) and each source

is matched to a distinct component with the Hungarian algorithm:

$$|\rho| = \begin{pmatrix} 0.0001 & \mathbf{0.9998} & 0.0223 \\ 0.0148 & 0.0182 & \mathbf{0.9997} \\ \mathbf{0.9996} & 0.0295 & 0.0011 \end{pmatrix}. \tag{11}$$

The matching is $s_1 \to$ IC1 (0.99975), $s_2 \to$ IC2 (0.99972) and $s_3 \to$ IC0 (0.99956): the order differs from the original and some signs are flipped, exactly as the ambiguities predict. Each row has one entry near one and two near zero, so the separation is clean.

4.3 Separation Quality and Stability

The mean matched absolute correlation is **0.99968**. The remaining gap of about 3×10^{-4} is plausibly due to the added noise, the finite sample size $N = 2000$, and the fact that the periodic waveforms are only approximately independent; algorithm convergence is not the cause, since it terminated cleanly after 7 iterations.

A single run should not, however, be trusted blindly. Repeating FastICA for seeds 0–9 gives a mean matched correlation of about 0.9997 for eight seeds but about 0.837 for seeds 4 and 5 (Table 3), meaning that one component was recovered poorly for those initialisations. This was not diagnosed further; a poor local solution of the non-convex contrast is the most likely explanation.

FastICA seeds	**Mean matched $\lvert\rho\rvert$**	**Interpretation**
0, 1, 2, 3, 6, 7, 8, 9	≈ 0.9997	clean separation
4, 5	≈ 0.837	one component poorly recovered

Table 3: Sensitivity of FastICA on the original sources to the random initialisation.

4.4 Why Correlation Rather Than MSE

ICA determines each source only up to an unknown scale and sign. Suppose a component is perfectly recovered but returned as $\hat{s} = \alpha s$ with $\alpha = -2$. Its mean squared error against s is $\mathbb{E}[(\hat{s} - s)^2] = (\alpha - 1)^2 \operatorname{Var}(s) = 9$, which wrongly signals a failure, whereas the absolute correlation is $|\rho(\alpha s, s)| = 1$ for every $\alpha \neq 0$. The absolute Pearson correlation is invariant to both scale and sign, which is why it is the appropriate metric.

5 PCA versus ICA: Decorrelation Is Not Independence

5.1 Why PCA Cannot Recover the Sources

Both methods are applied to the same mixtures. For unit-variance independent sources the covariance of the observations is

$$\Sigma_X = AA^\top = U \operatorname{diag}(\sigma_i^2)\, U^\top, \tag{12}$$

where U contains the left singular vectors of A. PCA returns the columns of U, which are *orthogonal* by construction, whereas the columns of A are generally *not* orthogonal. PCA therefore recovers the sources only in the special case that A has orthogonal columns. Here the PCA matched absolute correlations are 0.620, 0.671 and 0.816 with mean **0.702**, compared with **0.9997** for ICA. Because the first principal direction absorbs 10.2 of the total variance, each PC stays a mixture of sources, as Figure 3 (left) shows.

5.2 Both Methods Produce Uncorrelated Outputs

The sample covariance matrices of both output sets are diagonal to within $\sim 10^{-15}$. The PCA outputs have variances 10.242, 0.915 and 0.107 (the eigenvalues of Σ_X), and the ICA outputs have

variance 1.0005 each. The factor $1.0005 = 2000/1999$ is simply the ratio between the N-denominator used when `FastICA` scales its output and the $(N-1)$-denominator used for the covariance. Both outputs are uncorrelated, so a diagonal covariance matrix says nothing about independence.

5.3 A Higher-Order Dependence Measure

For output components z_i and z_j define the covariance of their squares,

$$D_{ij} = \mathbb{E}[z_i^2 z_j^2] - \mathbb{E}[z_i^2]\,\mathbb{E}[z_j^2], \qquad i \neq j. \tag{13}$$

If z_i and z_j are independent and the moments exist, then $D_{ij} = 0$, but a zero value alone does not prove independence. The diagonal entries are variances of squared components and are not expected to vanish. D is computed on unit-variance outputs so that scales are comparable:

$$D_{\text{PCA}} = \begin{pmatrix} 0.908 & -0.573 & -0.265 \\ -0.573 & 1.710 & -0.510 \\ -0.265 & -0.510 & 1.522 \end{pmatrix}, \qquad D_{\text{ICA}} = \begin{pmatrix} 0.811 & 0.0139 & 0.0080 \\ 0.0139 & 0.523 & 0.0155 \\ 0.0080 & 0.0155 & 0.0337 \end{pmatrix}. \tag{14}$$

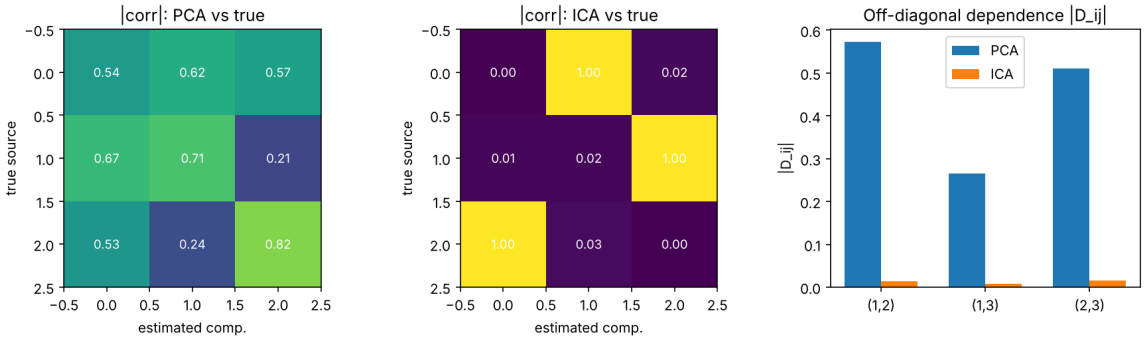


Figure 3: Absolute correlation with the true sources for PCA and ICA outputs, and the off-diagonal magnitudes $|D_{ij}|$.

Output set	**Matched mean $\lvert\rho\rvert$**	**Covariance diagonal?**	**Mean off-diag. $\lvert D_{ij}\rvert$**
True sources	1 (reference)	—	0.0065
PCA outputs	0.702	yes	0.450
ICA outputs	**0.9997**	yes	**0.0124**

Table 4: Second-order versus higher-order comparison of the two methods on the same observations.

5.4 Interpretation

The PCA outputs have large squared-component correlations (for example $D_{12} = -0.573$), so they are clearly statistically dependent even though they are uncorrelated. The ICA outputs sit near the level of the true sources themselves (0.0124 versus 0.0065). A small D_{ij} is necessary for independence but not sufficient, so this statistic supports the claim without proving it.

The distinction can be summarised by the chain *independent* $\Rightarrow$ *uncorrelated*, with no converse. A classic counterexample is $x \sim \mathcal{U}(-1, 1)$ and $y = x^2$: then $\text{Cov}(x, y) = \mathbb{E}[x^3] - \mathbb{E}[x]\mathbb{E}[x^2] = 0$ although y is a deterministic function of x. PCA enforces only decorrelation, a second-order property; independence requires the joint density to factorise and therefore constrains moments of all orders. The comparison of covariance (both diagonal) with D_{ij} (only ICA small) illustrates exactly this gap. FastICA does not minimise D_{ij} directly; it maximises non-Gaussianity, and the small D_{ij} is a by-product of recovering independent sources.

6 Gaussian Sources and Non-Identifiability

6.1 Random-Rotation Baseline

The sources are replaced by three i.i.d. standard Gaussian sequences (`default_rng(1)`, scaled to unit standard deviation) and mixed with the same matrix A. For three Gaussian sources a matched correlation need not be close to zero: matching the largest entries of a small correlation matrix produces a non-zero score by chance. To quantify this, X_{gauss} is centred and whitened and 200 random orthogonal matrices (QR factors of Gaussian matrices, `default_rng(123)`) are applied. For each rotation the mean matched absolute correlation with the true sources is computed. The baseline has mean 0.827, 5th percentile 0.694 and 95th percentile 0.955.

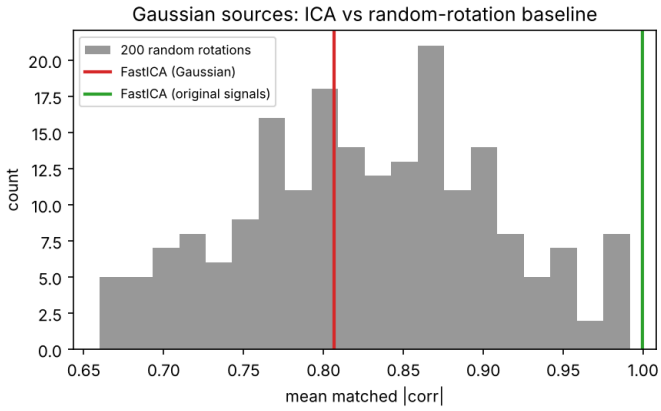


Figure 4: Gaussian case: distribution of matched correlation over 200 random rotations, with the FastICA score marked.

6.2 FastICA on Gaussian Sources

FastICA on X_{gauss} gives a mean matched absolute correlation of 0.807 after 12 iterations with no convergence warning, which lies inside the random-rotation baseline (Table 5). The corresponding score for the original illustrative signals is 0.99968. Across FastICA seeds 0–9 the Gaussian score varies between 0.806 and 0.884, so the answer depends on the initialisation, which is the behaviour expected when no rotation is preferred. Neither number was assumed in advance to fall inside or outside the baseline range.

Scenario	**Mean matched $\lvert\rho\rvert$**	**Iterations**
Random rotations of whitened Gaussian data	0.827 (5th–95th: 0.694–0.955)	—
FastICA on Gaussian sources	0.807	12
FastICA on original sources (Section 4)	0.99968	7

Table 5: Separation scores with and without the non-Gaussianity that ICA requires.

6.3 The Identifiability Condition

For i.i.d. Gaussian sources an arbitrary orthogonal rotation Q of the whitened data is again jointly Gaussian with identity covariance:

$$\mathbf{s} \sim \mathcal{N}(\mathbf{0}, I) \implies Q\mathbf{s} \sim \mathcal{N}(\mathbf{0}, QQ^\top) = \mathcal{N}(\mathbf{0}, I). \tag{15}$$

Thus $\mathbf{s}$ and $Q\mathbf{s}$ have exactly the same distribution, every rotation explains the observations equally well, and second-order statistics, which are all that whitened Gaussian data contain, cannot select

one of them. The sources are identifiable only up to an arbitrary rotation, that is, not identifiable. This is why the standard ICA model allows *at most one* Gaussian source.

6.4 Kurtosis as a Non-Gaussianity Measure

The excess kurtosis $\mathrm{kurt}(y) = \mathbb{E}[y^4]/\mathbb{E}[y^2]^2 - 3$ vanishes for a Gaussian. Table 6 lists the sample values together with the theoretical values: a sine has -1.5, a square wave -2 and a uniformly distributed sawtooth -1.2.

Source	Sample excess kurtosis	Theory
Sinusoid s_1	-1.479	-1.5
Square wave s_2	-1.968	-2.0
Sawtooth s_3	-1.191	-1.2
Gaussian sources	$-0.046,\ 0.082,\ 0.111$	0

Table 6: Excess kurtosis (`scipy.stats.kurtosis`, `fisher=True`) of the sources in both experiments.

Non-Gaussianity is what lets FastICA find the right directions. For independent unit-variance sources s_i and a unit-norm combination $y = \sum_i a_i s_i$ with $\sum_i a_i^2 = 1$, the additivity of cumulants gives

$$\mathrm{kurt}(y) = \sum_i a_i^4 \,\mathrm{kurt}(s_i), \qquad \text{so} \qquad |\,\mathrm{kurt}(y)| \le \max_i |\,\mathrm{kurt}(s_i)|, \tag{16}$$

with equality only when $\mathbf{a}$ is a coordinate vector. A rotation that mixes the sources pushes the kurtosis of each output toward zero, so maximising $|\,\mathrm{kurt}\,|$ (or negentropy) singles out the true directions. The sample values deviate from theory because of finite N: the standard error of the sample kurtosis of Gaussian data is approximately $\sqrt{24/N} \approx 0.11$ for $N = 2000$, which matches the observed deviations of ± 0.1, and the added noise slightly shifts the non-Gaussian cases.

Part II

PCA for Dimensionality Reduction

7 Modulation Dataset and Feature Extraction

7.1 Synthetic Modulation Dataset

Parts II–IV use a synthetic automatic modulation classification (AMC) dataset with four digital schemes observed in additive white Gaussian noise with an unknown carrier phase. For each class, 250 signals of 256 complex symbols are generated, giving 1000 signals in total at an SNR of 20 dB (the synthetic dataset is used, not `load_digits`, because it supports the SNR sweep of Section 14). Each signal is multiplied by a random carrier phase $e^{j\varphi}$ with $\varphi \sim \mathcal{U}(0, 2\pi)$, and complex Gaussian noise is added with variance set from the signal power P:

$$\sigma_n^2 = \frac{P}{10^{\mathrm{SNR}/10}}, \qquad n_k \sim \mathcal{CN}(0, \sigma_n^2). \tag{17}$$

The four constellations are BPSK (± 1), QPSK ($e^{j(\pi/4 + m\pi/2)}$), 8PSK ($e^{jm\pi/4}$) and 16QAM (levels $\{-3, -1, 1, 3\}$ per axis, normalised by $\sqrt{10}$). The unknown carrier phase is a nuisance parameter; features are chosen to be invariant to it.

7.2 The Twelve Statistical Features

Each received signal r is normalised to unit average power. With $a = |r|$ and $\varphi = \angle r$, the second-order moments and fourth-order cumulants are

$$C_{20} = \mathbb{E}[r^2], \qquad C_{21} = \mathbb{E}[|r|^2], \tag{18}$$

$$C_{40} = \mathbb{E}[r^4] - 3C_{20}^2, \qquad C_{42} = \mathbb{E}[|r|^4] - |C_{20}|^2 - 2C_{21}^2. \tag{19}$$

The twelve features are listed in Table 7.

#	Feature	Definition
1	std$\|r\|$	standard deviation of the envelope a
2	mean$\|r\|^2$	mean of a^2 (constant after power normalisation)
3	std$\|r\|^2$	standard deviation of a^2
4	kurt$\|r\|$	fourth standardised moment of a
5	skew$\|r\|$	third standardised moment of a
6	$\|C_{20}\|$	magnitude of the second-order moment
7	$\|C_{40}\|$	magnitude of the fourth-order cumulant
8	$\|C_{42}\|$	magnitude of the conjugate fourth-order cumulant
9	$\|C_{40}\|/C_{21}^2$	normalised C_{40} (duplicates feature 7 since $C_{21} = 1$)
10	$\|C_{42}\|/C_{21}^2$	normalised C_{42} (duplicates feature 8)
11	std(φ)	standard deviation of the instantaneous phase
12	mean$\|\varphi\|$	mean absolute phase

Table 7: The twelve amplitude-moment and cumulant features used for Parts II–IV.

7.3 Theoretical and Measured Class Signatures

For noiseless unit-power constellations the cumulant magnitudes are known in closed form, and they are what the features are designed to exploit. The measured class means at 20 dB agree closely with theory (Table 8).

	Theory (noiseless)			**Measured mean at 20 dB**		
Class	$\|C_{20}\|$	$\|C_{40}\|$	$\|C_{42}\|$	$\|C_{20}\|$	$\|C_{40}\|$	$\|C_{42}\|$
BPSK	1	2	2	0.990	1.961	1.961
QPSK	0	1	1	0.051	0.969	0.984
8PSK	0	0	1	0.052	0.060	0.984
16QAM	0	0.68	0.68	0.065	0.660	0.669

Table 8: Cumulant magnitudes: theoretical constellation values versus measured class means.

Two further features separate the families: the envelope spread std$|r|$ is about 0.070 for the three constant-envelope PSK classes (caused only by noise) but 0.326 for 16QAM, which has several amplitude levels; and the phase spread std(φ) grows from 1.592 for BPSK to about 1.77–1.80 for the higher-order constellations.

7.4 Redundancy Among the Features

The twelve features are far from independent. After unit-power normalisation mean$|r|^2 \equiv 1$ is constant, and because $C_{21} = 1$ the normalised cumulant ratios (features 9 and 10) duplicate features 7 and 8. Only about seven directions therefore carry variance, a fact that dominates the PCA results below.

8 Standardisation and Intrinsic Dimensionality

8.1 Why Standardise

Each feature is transformed to

$$z_{nm} = \frac{x_{nm} - \bar{x}_m}{s_m} \tag{20}$$

before PCA (`StandardScaler`). PCA maximises variance, so a feature measured on a larger numerical scale would dominate the leading component regardless of its information content; standardisation gives every feature equal weight. The constant feature mean$|r|^2$ has zero variance and is mapped to zero by `StandardScaler`, so the total variance of the standardised data is 11 rather than 12.

8.2 Scree Analysis

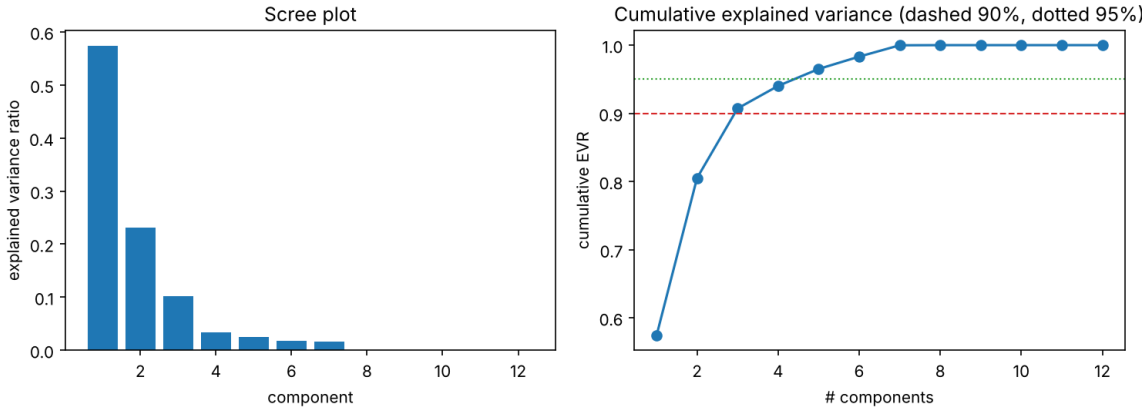


Figure 5: Scree plot (left) and cumulative explained variance ratio (right) of the standardised features.

Component	Eigenvalue λ_i	Explained ratio	Cumulative
1	6.3183	0.574	0.574
2	2.5464	0.231	0.805
3	1.1239	0.102	**0.907**
4	0.3640	0.033	0.940
5	0.2735	0.025	**0.965**
6	0.1996	0.018	0.983
7	0.1831	0.017	1.000
8	0.0023	0.000	1.000

Table 9: Principal component spectrum. Components 9–12 have eigenvalues below 10^{-3} and are omitted.

Table 9 and Figure 5 give the explained variance ratios 0.574, 0.231, 0.102, 0.033, 0.025, 0.018, 0.017 and approximately 0 for the rest. The cumulative ratio is 80.5% at $q = 2$, 90.7% at $q = 3$, 94.0% at $q = 4$ and 96.5% at $q = 5$. The minimum number of components is therefore **3 for 90%** and **5 for 95%** of the variance. The eigenvalues sum to $11.011 = 11 \times 1000/999$, the factor $1000/999$ arising because `StandardScaler` divides by N while the PCA covariance divides by $N - 1$; the ratios are unaffected.

8.3 Intrinsic Dimensionality Versus the Number of Classes

The dataset has four classes, yet three components already retain 90% of the variance. This is not a contradiction and one should not expect the two quantities to coincide. The number of classes is the number of groups, whereas the intrinsic dimensionality is the number of directions along which the features vary. Several classes can differ along only a few shared axes (for example, one axis separating BPSK from the other schemes), and within-class variation caused by noise and carrier phase contributes variance unrelated to class identity. Intrinsic dimension and class count are different quantities that merely happen to be of similar size here.

9 Reconstruction Error and Least-Squares Optimality

9.1 Derivation

PCA with q components gives the best rank-q linear approximation of the centred data in the least-squares sense. Writing $V_q \in \mathbb{R}^{12\times q}$ for the matrix of the first q eigenvectors, the reconstruction is $\hat{\mathbf{x}}_n^{(q)} = V_q V_q^\top \mathbf{x}_n$ and the quantity to be minimised is

$$\mathrm{MSE}(q) = \frac{1}{N}\sum_{n=1}^{N} \|\mathbf{x}_n - \hat{\mathbf{x}}_n^{(q)}\|_2^2 = \frac{1}{N}\|X - XV_qV_q^\top\|_F^2. \tag{21}$$

Because V_q has orthonormal columns, $\|X - XV_qV_q^\top\|_F^2 = \|X\|_F^2 - \mathrm{tr}(V_q^\top X^\top X V_q)$, so minimising the error is the same as maximising the retained variance $\mathrm{tr}(V_q^\top \Sigma V_q)$. The maximiser is the set of top-q eigenvectors, and the residual consists of the discarded eigenvalues. With the covariance denominator $N-1$,

$$\mathrm{MSE}(q) = \frac{N-1}{N}\sum_{i=q+1}^{12} \lambda_i. \tag{22}$$

For $q = 0$ this gives $\mathrm{MSE}(0) = \frac{999}{1000}\times 11.011 = 11.000$, which is exactly the mean squared norm of a standardised sample.

9.2 Numerical Verification

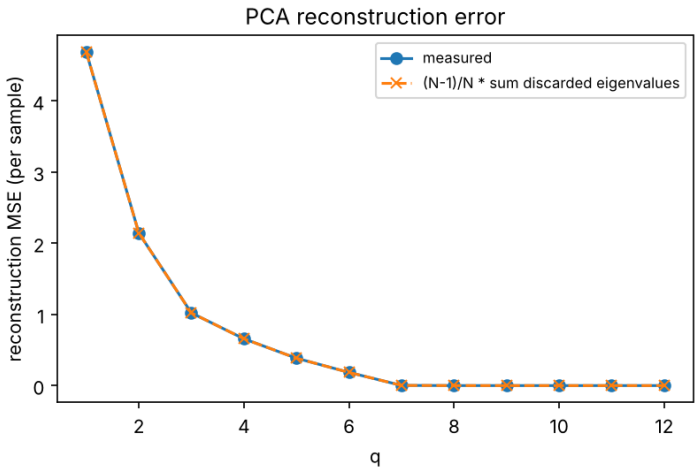


Figure 6: Reconstruction MSE as a function of the number of retained components q.

q	**Measured MSE**	$\frac{N-1}{N}\sum_{i>q}\lambda_i$
1	4.688053	4.688053
2	2.144244	2.144244
3	1.021504	1.021504
4	0.657865	0.657865
5	0.384658	0.384658
6	0.185291	0.185291
7	0.002401	0.002401
8	0.000103	0.000103

Table 10: Measured reconstruction error (via `inverse_transform`) against the eigenvalue formula.

The measured and predicted columns of Table 10 agree to $\sim 10^{-15}$ for every q (the full table is stored in `results/tables/B7_reconstruction_mse.csv`). Reconstruction error is therefore exactly the variance in the discarded directions. Since the eigenvectors capture the largest variances first, no other rank-q linear projection can achieve a lower least-squares error, which is the Eckart–Young theorem. The sharp drop between $q = 6$ and $q = 7$ ($0.185 \to 0.0024$) reflects the effective rank of about seven noted in Section 7.

10 Two-Dimensional Projection and Loading Analysis

10.1 Scatter Plot of the First Two Components

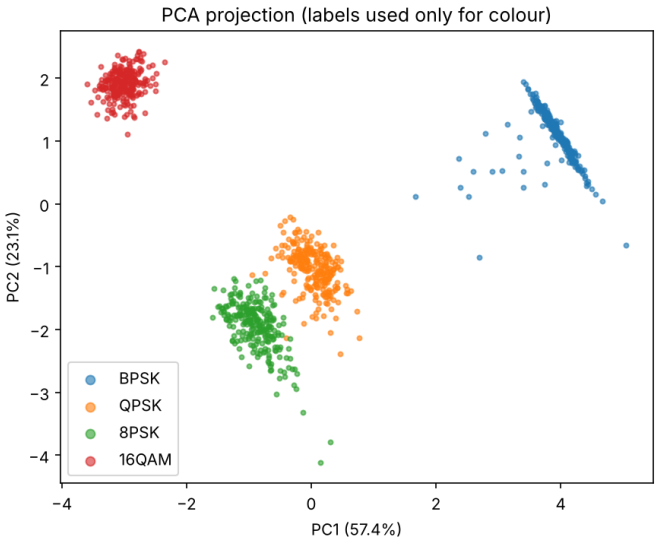


Figure 7: Standardised data projected onto the first two principal components. **Labels are used only for colouring; PCA was fitted without them.**

The standardised data are projected onto the first two principal components and coloured by modulation label (Figure 7). The labels were used only for colouring the plot and were not used to fit PCA. BPSK and 16QAM are well separated from every other class. QPSK and 8PSK are adjacent: measured in units of the pooled per-axis within-class standard deviation, their centroid distance is only about 3.9, against at least 12.5 for every other pair (Table 11), and their tails overlap slightly. Two components hold only 80.5% of the variance, so discriminative information in the remaining 19.5% is discarded, and a low-variance direction can still be the discriminative one.

Pair	BPSK–QPSK	BPSK–8PSK	BPSK–16QAM	QPSK–8PSK	QPSK–16QAM	8PSK–16QAM
Separation	12.5	15.3	22.3	**3.9**	16.0	15.2

Table 11: Centroid distance in the two-component PCA plane, in units of the pooled per-axis within-class standard deviation.

10.2 Loading Vectors

The loadings of the first three components are listed in Table 12. The largest absolute loadings on PC1 are $|C_{42}|$ (0.386), $|C_{20}|$ (0.352) and $|C_{40}|$ (0.322); the normalised cumulant ratios duplicate the last and first of these.

Feature	**PC1**	**PC2**	**PC3**
std$\lvert r\rvert$	-0.275	$+0.435$	-0.001
mean$\lvert r\rvert^2$	0.000	0.000	0.000
std$\lvert r\rvert^2$	-0.275	$+0.434$	-0.001
kurt$\lvert r\rvert$	$+0.243$	-0.395	$+0.011$
skew$\lvert r\rvert$	$+0.248$	-0.374	0.000
$\lvert C_{20}\rvert$	$+0.352$	$+0.252$	$+0.066$
$\lvert C_{40}\rvert$	$+0.322$	$+0.322$	$+0.084$
$\lvert C_{42}\rvert$	$+0.386$	$+0.090$	$+0.058$
$\lvert C_{40}\rvert/C_{21}^2$	$+0.322$	$+0.322$	$+0.084$
$\lvert C_{42}\rvert/C_{21}^2$	$+0.386$	$+0.090$	$+0.058$
std(φ)	-0.314	-0.198	$+0.335$
mean$\lvert\varphi\rvert$	-0.022	-0.010	$+0.929$

Table 12: Loading vectors of the first three principal components (rows of `pca.components_`).

These loadings make physical sense. $|C_{20}| \approx 1$ only for the real-valued BPSK and ≈ 0 for the others, and $|C_{40}|$ takes the values $2, 1, 0, 0.68$ and $|C_{42}|$ the values $2, 1, 1, 0.68$ for BPSK, QPSK, 8PSK and 16QAM (Table 8), so PC1 is largely a *constellation-order* axis with BPSK far to the right. The PSK-versus-QAM split is carried by PC2, whose largest loadings are the amplitude-spread features std$|r|$ and std$|r|^2$ ($\approx +0.43$) together with the opposite-signed envelope shape features: PSK has a constant envelope while 16QAM has several amplitude levels. PC3 is dominated by mean$|\varphi|$ (0.929), a feature whose class means all equal approximately $\pi/2$ (between 1.565 and 1.571); this component therefore mostly captures nuisance variation caused by the unknown carrier phase, not class information.

Part III

k-Means Clustering and Cluster Validity

11 Clustering at $k = 4$

11.1 The k-Means Objective

k-means partitions the standardised feature vectors into k clusters by minimising the within-cluster sum of squares (the *inertia*),

$$J = \sum_{c=1}^{k} \sum_{\mathbf{x}_i \in \mathcal{C}_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|_2^2, \tag{23}$$

using Lloyd's algorithm, which alternates between assigning each point to its nearest centroid and recomputing each centroid as the mean of its members. J never increases, so the procedure converges, but only to a local minimum; `n_init=10` restarts from different initialisations and keeps the best, and `random_state=0` fixes them.

11.2 Internal and External Validity

Two families of cluster-validity indices are used for different purposes (Table 13).

	Internal	**External**
Requires labels?	No	Yes
Examples	Inertia, silhouette score	ARI, NMI
Purpose	*Selecting* k	*Evaluating* the result

Table 13: Internal versus external clustering validity.

The silhouette of a sample i compares its mean distance $a(i)$ to the other members of its own cluster with the smallest mean distance $b(i)$ to any other cluster:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1]. \tag{24}$$

The Rand index is the fraction of sample pairs on which two partitions agree (both together or both apart). The *adjusted* Rand index corrects for chance:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{\max(\mathrm{RI}) - \mathbb{E}[\mathrm{RI}]}, \tag{25}$$

so that 1 means perfect agreement and a random assignment scores ≈ 0. The score of one random assignment is a random variable with mean ≈ 0 but non-zero spread: over 200 random four-way assignments of this dataset the mean ARI was -5×10^{-5} with standard deviation 0.0013, so a single draw need not be exactly zero. NMI measures the mutual information between the two partitions, normalised to $[0, 1]$.

11.3 Results at $k = 4$

Running `KMeans(n_clusters=4, n_init=10, random_state=0)` on the standardised 12-dimensional features gives an inertia of 2088.2 and a silhouette score of 0.480. Evaluated against the true labels (used here only for evaluation), the ARI is 0.9973 and the NMI 0.9953. The contingency table is Table 14.

True class	**c0**	**c1**	**c2**	**c3**
BPSK	0	249	0	1
QPSK	0	0	0	250
8PSK	250	0	0	0
16QAM	0	0	250	0

Table 14: Contingency table of true class (rows) against k-means cluster index (columns) at $k = 4$.

Exactly one sample, a single BPSK signal, is misassigned. Cluster indices are arbitrary labels assigned by the initialisation, so comparing them with class labels using ordinary accuracy would depend on a meaningless permutation unless the clusters were first optimally matched to classes. ARI and NMI are permutation-invariant and avoid this problem.

12 Choosing k Without Labels

12.1 The Inertia and Silhouette Sweep

For $k = 2, \ldots, 8$ the inertia and silhouette score are recorded. The ARI is computed for evaluation only and is not used to select k (Table 15, Figure 8).

k	Inertia	Silhouette	ARI (eval only)
2	5581.4	0.509	0.333
3	2531.9	**0.588**	0.714
4	2088.2	0.480	**0.997**
5	1868.4	0.467	0.877
6	1672.0	0.399	0.786
7	1505.7	0.390	0.711
8	1352.7	0.394	0.699

Table 15: Inertia, silhouette score and (evaluation-only) ARI as a function of k.

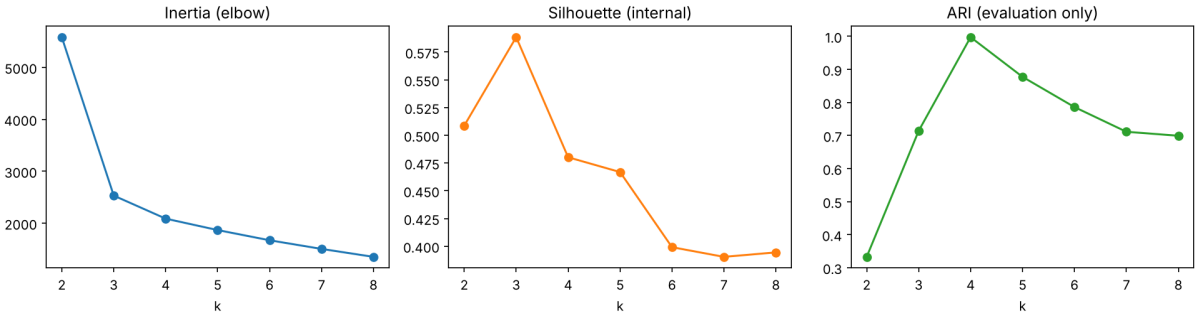


Figure 8: Inertia (elbow plot), silhouette score and evaluation-only ARI versus k.

12.2 Selection Using Internal Criteria Only

Using only internal criteria, as one would have to in a real unsupervised problem, the choice is $\mathbf{k = 3}$. The silhouette is highest there (0.588), and the elbow, measured as the largest second difference of the inertia (2606 at $k = 3$ against 224 at $k = 4$), is also at 3. The bend at $k = 4$ is weaker but still visible, so the elbow plot alone is not decisive.

12.3 Where the Internal Index Fails

The highest silhouette is at $k = 3$ but the highest ARI is at $k = 4$: the two criteria disagree. Inspecting the $k = 3$ partition (Table 16) shows that it merges QPSK and 8PSK into a single cluster while BPSK and 16QAM stay separate, exactly the close pair seen in the PCA scatter plot. The silhouette measures compactness relative to separation, and merging two nearby classes yields fewer, compact and well-separated clusters, so it rewards the merge even though a true class is lost.

True class	c0	c1	c2
BPSK	0	250	0
QPSK	250	0	0
8PSK	250	0	0
16QAM	0	0	250

Table 16: Contingency table at $k = 3$: QPSK and 8PSK share one cluster (ARI = 0.714).

12.4 Practical Safeguards

Without labels, a single internal index can be systematically wrong when classes lie close together. A practical safeguard is to combine several indices (silhouette, Davies–Bouldin, Calinski–Harabasz, the gap statistic), to check stability across resamples and random seeds, and to inspect the re-

sulting clusters with domain knowledge, here the knowledge that QPSK and 8PSK are distinct constellations.

13 PCA Before Clustering

The data are clustered in two ways, using $k = 4$ in both: on the full 12-dimensional standardised features and on the two-dimensional PCA representation of Section 10 (Table 17).

Space	ARI	NMI	Silhouette
12-D standardised	0.9973	0.9953	0.480
2-D PCA	0.9867	0.9805	0.744

Table 17: k-means ($k = 4$) in the full feature space versus the two-component PCA plane.

Reducing the dimensionality did not improve the ARI; it dropped slightly from 0.997 to 0.987, a difference of a handful of samples, because the 2-D projection discards the 19.5% of the variance that helps separate QPSK from 8PSK. The silhouette nevertheless jumps from 0.480 to 0.744, because distances are measured in a different space: the 12-D space contains redundant and noisy dimensions that add within-cluster spread. Scoring the 12-D clustering inside the 2-D space already gives 0.742, so almost all of the gain comes from the change of space, not from a better clustering. Silhouette values computed in different feature spaces are therefore not comparable.

14 Clustering Performance versus SNR

14.1 Protocol

The modulation dataset is regenerated at SNR $\in \{20, 15, 10, 5, 0\}$ dB. At each SNR the complete pipeline is repeated: feature extraction, standardisation and k-means with $k = 4$. Because the single seed-0 dataset can be lucky or unlucky, the experiment is also repeated over 10 independently generated datasets (seeds 0–9) and summarised as mean ± standard deviation.

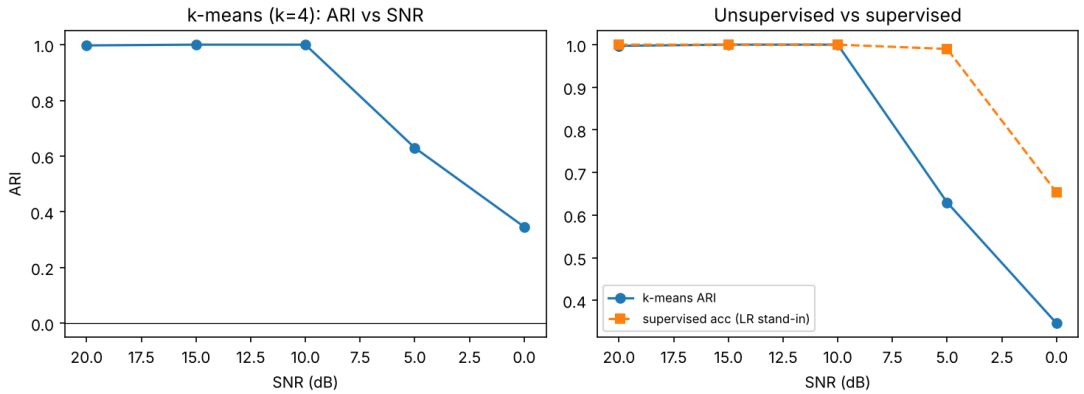


Figure 9: k-means ARI versus SNR (seed-0 dataset and ten-dataset average) together with the supervised stand-in accuracy.

SNR (dB)	ARI (seed 0)	ARI, 10 datasets	NMI (seed 0)	Supervised acc.
20	0.997	0.9997 ± 0.0008	0.995	1.00
15	1.000	0.967 ± 0.103	1.000	1.00
10	1.000	0.799 ± 0.173	1.000	1.00
5	0.630	0.620 ± 0.009	0.771	0.99
0	0.346	0.321 ± 0.018	0.409	0.65

Table 18: Clustering quality and the supervised stand-in as a function of SNR.

14.2 Where Clustering Deteriorates

The single seed-0 curve is flat down to 10 dB and then drops sharply, but that is partly luck. Averaged over ten datasets the degradation begins earlier (0.97 at 15 dB and 0.80 ± 0.17 at 10 dB) and is large at 5 dB (0.62). Random clustering gives an ARI of about 0 (here $-5 \times 10^{-5} \pm 0.0013$), so at 0 dB the ARI of 0.32–0.35 is still clearly above chance but far from useful.

14.3 Comparison with Supervised Classification

The Lab 3 accuracy curve is not part of this repository, so the supervised column of Table 18 is a stand-in: a standardised logistic regression on the same features with a 70/30 stratified split (seed 0). It should be replaced with the Lab 3 accuracies if these differ. With this stand-in, clustering falls behind from about 10 dB (dataset-averaged) or 5 dB (seed 0), where supervised accuracy is still ≥ 0.99. A supervised model can place decision boundaries between classes whose features are shifted by noise but still consistently different, even when the clusters overlap, whereas k-means needs compact and well-separated groups to exist in the first place. At 0 dB even the supervised stand-in drops to 0.65.

Part IV

When k-Means Is the Wrong Tool

15 Comparing Clustering Algorithms

15.1 Setup

At 20 dB and $k = 4$, three algorithms are applied to the same standardised 12-dimensional features and scored against the true labels: `KMeans(n_init=10, random_state=0)`, `GaussianMixture(covariance_type="full", random_state=0)` (default `n_init=1`, `reg_covar=`10^{-6}) and `AgglomerativeClustering(linkage="ward")`, which is deterministic. A Gaussian mixture models the density as $p(\mathbf{x}) = \sum_c \pi_c \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_c, \Sigma_c)$ and is fitted by expectation–maximisation, alternating soft responsibilities $\gamma_{ic} \propto \pi_c \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_c, \Sigma_c)$ with weighted updates of π_c, $\boldsymbol{\mu}_c$ and Σ_c. Ward linkage repeatedly merges the two clusters whose union produces the smallest increase in within-cluster sum of squares.

15.2 Results

Method	ARI	NMI	ARI, 10 datasets	NMI, 10 datasets
KMeans	0.9973	0.9953	0.9997 ± 0.0008	0.9995 ± 0.0015
GMM (full)	1.0000	1.0000	0.905 ± 0.152	0.932 ± 0.109
Agglomerative (ward)	1.0000	1.0000	0.978 ± 0.048	0.982 ± 0.039

Table 19: Algorithm comparison at 20 dB, $k = 4$, standardised 12-D features (single run and ten-dataset summary).

15.3 Interpretation

In the one required run, GMM and Ward both reach 1.0 while k-means reaches 0.9973, but this gap is a single sample (one BPSK point) and does not support any ranking. A single run reflects one dataset draw and one initialisation, so it cannot establish that one algorithm is generally better than another. The supplementary ten-dataset experiment (seeds 0–9, with the dataset seed also used as the model seed) reverses the ranking: k-means is best and very stable, Ward is next, and the full-covariance GMM is the worst and most variable. A likely but untested reason is that the full covariance of the GMM is estimated on features that are nearly collinear or duplicated (Section 7), which makes some component covariances almost singular, combined with a single initialisation.

The structural difference between the methods is that k-means favours compact, approximately spherical clusters of comparable size, because it assigns each point to its nearest centroid by squared Euclidean distance. A GMM relaxes this by giving every cluster its own full covariance matrix, so that clusters can be elliptical with different orientations and sizes, and by using soft assignments with mixture weights instead of hard nearest-centroid assignment.

16 Failure Case: Two Moons

16.1 k-Means on Non-Convex Clusters

To construct a failure case deliberately, `make_moons(n_samples=500, noise=0.05, random_state=0)` generates two interleaved crescents, and k-means with $k = 2$ is applied (Figure 10).

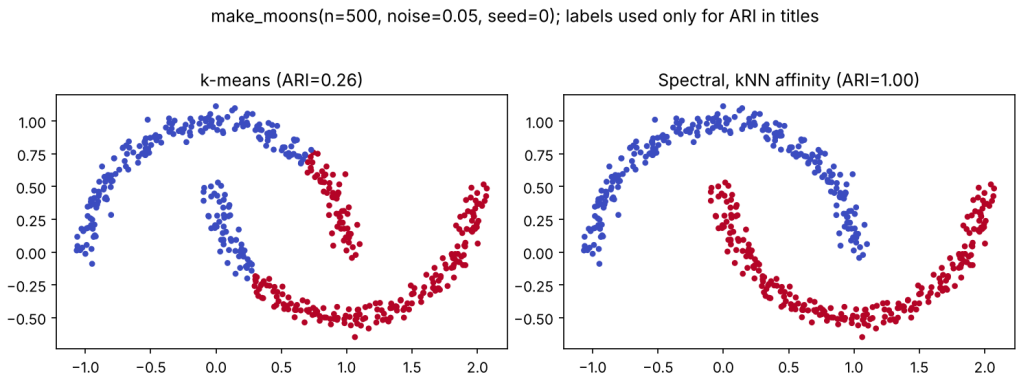


Figure 10: k-means (left) versus spectral clustering (right) on `make_moons(500, noise=0.05)`.

k-means obtains ARI = 0.257 and NMI = 0.195: it cuts the data with a straight line. The reason follows from the decision rule. A point is assigned to centroid $\boldsymbol{\mu}_1$ rather than $\boldsymbol{\mu}_2$ when $\|\mathbf{x} - \boldsymbol{\mu}_1\|^2 < \|\mathbf{x} - \boldsymbol{\mu}_2\|^2$, and the boundary is the set where equality holds,

$$(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^\top \mathbf{x} = \tfrac{1}{2}(\|\boldsymbol{\mu}_2\|^2 - \|\boldsymbol{\mu}_1\|^2), \tag{26}$$

which is the *perpendicular bisector* of the segment joining the centroids and hence a straight line in two dimensions. Each crescent contains points on both sides of any straight line through the middle of the data, so no linear boundary can separate the two non-convex clusters.

16.2 Spectral Clustering

`SpectralClustering(n_clusters=2, affinity="nearest_neighbors", random_state=0)` on the same data gives ARI = 1.0 and NMI = 1.0. It builds a nearest-neighbour graph with affinity matrix W, forms the graph Laplacian $L = D - W$ with degree matrix D, and splits the graph using the leading non-trivial eigenvectors of L. Points are linked to nearby points, so each crescent forms a connected chain and cluster membership follows connectivity rather than distance to a centre. `scikit-learn` warned that the graph was not fully connected; the result was nevertheless perfect here.

17 Conclusion and Summary

This laboratory investigation examined three unsupervised techniques and the conditions under which each succeeds or fails:

1. **Blind Source Separation:** FastICA recovered three mixed sources with a mean matched correlation of 0.99968 against 0.702 for PCA. PCA outputs are uncorrelated but dependent (mean off-diagonal $|D_{ij}| = 0.450$), whereas ICA outputs reach the level of the true sources (0.0124 versus 0.0065), demonstrating that decorrelation is weaker than independence. A seed check showed that a single FastICA run can still land in a poor solution (≈ 0.837 for two of ten seeds).
2. **Identifiability:** With Gaussian sources FastICA scored 0.807, inside the random-rotation baseline (0.827, 5th–95th percentile 0.694–0.955), confirming that at most one Gaussian source is allowed. Kurtosis explains why non-Gaussian mixtures can be unmixed.
3. **Dimensionality Reduction:**
   - Three components retained 90% and five retained 95% of the variance of twelve highly redundant features.
   - Reconstruction error matched $\frac{N-1}{N}\sum_{i>q}\lambda_i$ to 10^{-15}, verifying least-squares optimality.
   - The two-component plane separates BPSK and 16QAM clearly but leaves QPSK and 8PSK close.
4. **Clustering and Validity:**
   - k-means at $k = 4$ gave ARI = 0.9973, but the silhouette and elbow selected $k = 3$ by merging QPSK and 8PSK, a documented failure of a single internal index.
   - PCA to two dimensions raised the silhouette ($0.480 \rightarrow 0.744$) without improving the ARI, so silhouettes from different spaces are not comparable.
   - Over ten datasets the ARI falls from 0.9997 at 20 dB to 0.80 at 10 dB and 0.32 at 0 dB.
5. **Choice of Algorithm:** Single-run rankings of k-means, GMM and Ward were unreliable and were reversed over ten datasets. On two moons k-means reached ARI = 0.257 where spectral clustering reached 1.0, because a centroid rule can only draw straight boundaries.